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TransCelerate DDF Commercial Adaptation POC: Status and Next Steps

Current Status

Overview

This document provides an assessment of the current status of the TransCelerate DDF Commercial Adaptation Proof of Concept (POC) and outlines the next steps to complete the POC in alignment with the defined POC strategy.

Implementation Status

1. SDR Core API

- **Status:** Basic implementation in place
- **Completed:**
 - Basic project structure set up
 - USDM data model implemented (StudyDefinition.cs)
 - CRUD API endpoints implemented (StudyDefinitionsController.cs)
 - MongoDB integration for data storage (StudyDefinitionService.cs)
 - Basic configuration (appsettings.json)
 - Swagger documentation
- **Pending:**
 - Authentication and authorization
 - Data validation
 - Error handling improvements
 - Metrics collection
 - Comprehensive testing

2. StudyBuilder Adapter

- **Status:** Initial structure only
- **Completed:**
 - Basic project structure set up
 - README with requirements
- **Pending:**
 - Integration with study design systems
 - Transformation to USDM format
 - Validation against USDM schema
 - Submission to SDR Core API
 - Testing and metrics

3. EDC Adapter

- **Status:** Initial structure only
- **Completed:**
 - Basic project structure set up
 - README with requirements
- **Pending:**
 - Integration with EDC systems
 - Transformation from USDM to EDC-specific format
 - Form generation

- Edit check creation
- Testing and metrics

4. Biomedical Concepts

- **Status:** Initial set defined
- **Completed:**
 - Basic set of vital signs concepts
 - Pain assessment concept
- **Pending:**
 - Laboratory category concepts
 - Additional patient reported outcomes
 - Safety assessments
 - Comprehensive testing

5. Docker and Deployment

- **Status:** Basic setup in place
- **Completed:**
 - Docker files for each component
 - Docker Compose configuration
- **Pending:**
 - Testing of Docker deployment
 - Documentation of deployment process
 - Performance testing

Alignment with POC Strategy

POC Objectives

1. Validate Technical Feasibility

- **Status:** Partially implemented
- **Next Steps:** Complete the implementation of all components and test end-to-end workflow

2. Demonstrate Business Value

- **Status:** Not started
- **Next Steps:** Implement metrics collection and analysis

3. Identify Implementation Challenges

- **Status:** In progress
- **Next Steps:** Document challenges encountered during implementation

4. Refine Implementation Approach

- **Status:** Not started
- **Next Steps:** Analyze implementation experience and refine approach

POC Scope

- **In Scope Items Status:**
 - Core SDR deployment: Partially implemented
 - Integration with study design system: Not started
 - Integration with EDC system: Not started
 - Implementation of limited BiomedicalConcept set: Partially implemented
 - End-to-end workflow: Not started
 - Basic metrics collection: Not started

Next Steps to Complete POC

Phase 1: Complete Core Components (2 weeks)

1. SDR Core API

- Implement authentication and authorization
- Add data validation
- Improve error handling
- Set up metrics collection

2. BiomedicalConcepts

- Implement remaining concept categories

- Test and validate concepts

Phase 2: Implement Adapters (3 weeks)

1. StudyBuilder Adapter

- Select study builder system for integration
- Implement integration adapter
- Develop USDM transformation
- Test with sample study designs

2. EDC Adapter

- Select EDC system for integration
- Implement EDC adapter
- Develop form generation
- Test with sample study designs

Phase 3: End-to-End Testing (2 weeks)

1. Integration Testing

- Test full workflow from study design to EDC
- Measure performance and accuracy
- Document issues and solutions

2. Metrics Collection

- Implement metrics for all components
- Collect and analyze metrics
- Compare with success criteria

Phase 4: Evaluation and Documentation (1 week)

1. POC Evaluation

- Analyze results against objectives

- Document challenges and solutions
- Prepare final report

2. Implementation Planning

- Refine technical architecture
- Develop scaling strategy
- Create implementation roadmap

Conclusion

The POC implementation is currently in the early stages, with basic structure in place but significant work remaining to complete all components and demonstrate end-to-end functionality. By following the next steps outlined above, we can complete the POC and achieve the objectives defined in the POC strategy.